

Hamdy Freeg

Giza, Egypt

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[LinkedIn Profile](#) — [GitHub Profile](#)

Professional Summary

Passionate Computer Science student with a solid mathematical background, focusing on Backend Development and Systems Engineering. I enjoy building core logic, understanding low-level network mechanics, and solving complex algorithmic problems. Actively seeking a Software Engineering internship to apply my technical skills, write clean code, and gain hands-on experience in a real-world production environment.

Education

El Shorouk Academy

[2025] – [2028]

B.Sc. in Computer Science

Cairo, Egypt

- **Current Grade:** Excellent
- **Focus Areas:** Data Structures & Algorithms, Object-Oriented Programming, Systems Engineering.

Cairo University

[2024] – [2025]

Studies in Mathematics (Transferred)

Giza, Egypt

- **Relevant Coursework:** Linear Algebra, Calculus, Probability & Statistics, Mathematical Logic.

Courses & Certifications

- **Mastering critical SKILLS in Data Structures using C++** – (*Dr. Mostafa Saad / udemy*)
- **Mastering critical SKILLS in Algorithms using C++** – (*Dr. Mostafa Saad / udemy*)
- **Mastering 4 critical SKILLS using C++** – (*Dr. Mostafa Saad / udemy*)
- **Java Development SE** – (*ITI Mahara-tech*)
- **From Nand to Tetris** – (*Coursera*)
- **Object Oriented Programming in Java** – (*University of California San Diego / Coursera*)
- **Introduction to Networking** – (*NVIDIA / Coursera*)
- **JavaScript - The Complete Guide (Beginner + Advanced)** – (*Maximilian / udemy*)
- **Build Responsive Real-World Websites with HTML and CSS** – (*Maximilian / udemy*)
- **Mathematics for Computer Science** – (*MIT Open CourseWare*)
- **CS 61a** – (*UC Berkley*)
- **Mathematics for machine learning** – (*deeplearning.ai*)
- **Database Design and Basic SQL in PostgreSQL** – (*University of Michigan*)

Technical Skills

- **Programming Languages:** C++, Java, JavaScript, TypeScript, Bash Scripting, Python, SQL.
- **Core Concepts:** Backend Development, Systems Engineering, Data Structures & Algorithms, OOP, Networking, DBMS, PostgreSQL.
- **Tools & Environments:** Linux, Git, GitHub, VS Code, IntelliJ, LaTeX.

Projects

- **DNS Server (C++):** Developed a raw UDP DNS server matching RFC 1035 standards using POSIX sockets. Engineered a custom serialization and deserialization engine to process packet headers, questions, and answers, successfully resolving complex DNS pointer compression.
- **ASKI System (Ask.fm Clone):** Built a platform utilizing core OOP principles and custom relational logic to handle user management, anonymous questions, and chronological thread tracking with persistent data structures.
- **Library Management System:** Designed and implemented a robust management system featuring user authentication, multi-role access control (Admin/User), and efficient inventory tracking with strict focus on system correctness.

- **Custom Assembler (Java):** Developed a functional compiler toolchain component along with a formatting-agnostic file comparison tool to master low-level data flows and structural OOP patterns.

Languages

- **English:** Professional
- **Arabic:** Native